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How can industrial architecture be made more sustainable in the UK?

In a world and country that is constantly evolving and adapting to environmental challenges, maintaining long-lasting, sustainable conditions has become a priority for the development of society. Industrial architecture, although always changing, remains a significant factor in global emissions; this research aims to deepen understanding of industrial architecture's impacts on sustainability. This EPQ will explore the concept of sustainability and how it applies specifically to industrial architecture through its economic, environmental and societal impacts. Identifying the weaknesses in industrial architecture will provide insight into materials that may cause damage to the sustainable society the UK strives to achieve. This EPQ will also include a research project of the Life-Cycle Assessments (LCAs) of concrete, brick, metal and glass as these are the most common materials handled within industrial architecture. A more refined understanding of these materials is essential for advancing sustainable practices and promoting industrial architecture toward innovative solutions. By exploring LCAs, this research will also analyze the long-term and short-term consequences of low sustainability and highlight the critical role industrial architecture plays in creating a carbon-neutral society.

What does sustainability mean within architecture?

Before critically analysing sustainability and its applications in industrial architecture, defining the term "sustainability" is essential to establish a clear understanding. As defined by [\(Baker Associates, 2022\)](#) "Sustainable architecture is also referred to as green architecture or environmental architecture. It challenges architects to produce smart designs and use available technologies to ensure that structures generate minimal harmful effects to the ecosystem and the communities." This definition of sustainability, primarily in the industrial setting, highlights the importance of innovation, which will be explored further throughout the chapters while linking back to the key idea of green architecture. Sustainability in architecture does not only refer to environmental elements, but also societal, economic and political aspects which will be discussed further in the context of industrial architecture, as these affect all aspects of the environment ranging from wildlife to urban cities, affecting our individual lives and impacting the future development of urban, rural and wild areas across the world. This fundamental term of sustainable architecture will shape the understanding of industrial architectural practices to promote more sustainable practices within the UK.

(Lee, 2009) stated that "as sustainable design and development have emerged as one of the most compelling in the architecture of our time, as well as in society and politics at large, it is important to explore the changes that have occurred in the architectural profession as a result". This evidence supports that sustainable architecture does contain relevance in politics and society and impacts these on a deeper level. The author also emphasises the importance of exploring the changes that have taken place through innovations of architectural practices caused by sustainable architecture.

Why is sustainability important for industrial architecture?

Understanding the importance of sustainable architecture is a vital stepping stone into the deeper analysis of the industry sector. Economics, politics, society and the environment are all key factors of architecture that impact the UK in different ways, all being improved by the innovation of architectural practices, however still unable to maintain a sustainable country. Stated by (M Moser Associates, 2024) "Sustainable architecture can help reduce energy consumption by incorporating energy-efficient features such as insulation, heating and cooling systems" Reducing energy consumption is an extreme benefit to the sustainability of architecture and provides economic and environmental bonuses through less money being spent on appliances and can go towards better projects that will benefit society. Through the reduction of appliances this will provide benefits to the surrounding environment through reduced indirect emissions caused from electricity generation.

Societal impacts are a major concern when it comes to creating sustainable architecture in the industrial setting. Society as a whole is impacted by poor sustainability due to promoting poor lifestyles and providing an unsafe, poorer quality of life. According to the research of (Edwards and Naboni, 2013) "Green buildings generally help to cement neighbourhoods and encourage healthy lifestyles. Hence, there are benefits to companies, to individuals and to communities." This evidence suggests that high-quality sustainability promotes a more safe society and a higher-functioning community that can engage with others, benefitting overall communities; this can cause a higher employment rate due to more sustainable industrial buildings being built in neighbouring communities. Improving this employment rate through the multiplier effect, will improve the general lives of people in local communities as they will have more motivation to take part in society while also reducing the rate of crimes because of people being able to support themselves. Sustainable architecture can also provide inspiration for society to improve their impacts on the environment, similar to promoting healthier lifestyles for example cities may increase recycling.

The UK economy is also impacted by the sustainability of architecture, which includes large businesses and the general public. In this section of the chapter a fundamental understanding of the benefits sustainable architecture has on the economy will be established to show how the UK can grow economically to create a stronger financial state within the country through the application of sustainable architecture practices. In industrial architecture there are many alternatives to reduce the operational costs of the building through the use of LED lights and insulation which reduces the heat loss in the building, this means that less money will be spent through heating the structure. However in the modern day business market, less CEOs are attracted to sustainable architecture. According to (Babcock, 2016) "If the operational costs and up-front costs can be decreased, sustainable build may become a more attractive business venture" this research from Babcock is essential to understanding the mind of most business owners, large-scale or small-scale and states that the current issue with the promotion of sustainable architecture is the initial cost of the building compared to traditional due to renewable energy sources such as solar panels taking up extra space on the land which accounts to the total cost of the whole building. However, to allow this higher initial cost to be worth it long-term, renewable energy sources have benefits for the cost such as a longer lifetime compared to non-renewable sources of energy such as fossil fuels. This will allow, in the future, the sustainable building to last longer; therefore generating more money within its lifespan benefiting the economy in the end. Other factors are taken into consideration when purchasing industrial architecture such as operational costs which can be decreased, as discussed previously, which can make sustainable architecture more appealing to businesses. These factors that prevent the widespread construction and investment of sustainable architecture can be overcome to promote sustainability into different areas of architecture than just industrial such as civil architecture. Other ways we can promote sustainable architecture to business owners in an attempt to encourage sustainable practices is through the reduction of energy consumption, this can be achieved through energy efficient variations of household appliances that, on average, use the most energy. (Mariotti, 2023) states that "Green

building can reduce energy consumption by roughly 30%-40% compared to traditional building" this research offers an immense decrease in energy consumption to almost 50% which has benefits, for example: saving money. This benefit provides a stronger economy and can support the reduction of future inflation along with providing more opportunities for new homeowners to be able to support themselves and potential families. The money saved can also be explained through the reduction of operational costs which can also provide similar benefits through saving money. Reducing the energy consumption in industrial buildings is also essential to free up resources and energy which can be directed to other areas of the UK, possibly increasing energy security. Through sustainable industrial architecture, future employment opportunities can arise due to the promotion of sustainable architecture because more business owners will invest in this form of architecture to provide new careers in the industrial sector. With new employment in the industrial setting the upkeep of these buildings can be maintained which will benefit the financial state of the UK by increasing the industrial productivity and power of the country will increase foreign direct investments.

Sustainability is important within industrial architecture to assist the environment in its growth rather than its destruction. (Mariotti, 2023) refers to the differences between energy consumption of traditional versus sustainable buildings, "Green building can reduce energy consumption by roughly 30%-40% compared to traditional building" the benefits of this energy consumption decrease is the prevention of unneeded pollution through non-renewable energy sources such as fossil fuels being burnt and causing damaging effects to the environment by polluting the surrounding air. Through sustainable architecture consequences such as pollution can be exterminated by reducing the energy used during industrial architecture. Preservation is another growing concern in the environment which can be achieved through sustainable architecture with long-lasting buildings and joint efforts between companies to support this. (James, 2015) mentions "Companies of all sizes embrace sustainability in an effort to preserve resources for future generations while continuing to create value for current generations" this supports the idea that current companies are in contact to preserve the environment for those in the future this includes attempts to reduce global warming worldwide by making efforts within the UK. The impact of global warming can be reduced through adaptation and mitigation strategies such as new architectural practices and risk management by regular reviews and addressing new challenges within industrial architecture.

Political Policies towards sustainability

Before understanding the political policies the UK has set in place to achieve sustainability, first the term net-zero must be defined as it is essential to understand this terminology to then analyse the political changes in the UK. According to (UK Government, 2019) "Net-zero greenhouse gas emissions means that any residual emissions are balanced by an equivalent amount of greenhouse gas removal from the atmosphere." The term residual emissions refers to industries that may produce unavoidable emissions, which are cancelled out by greenhouse gas removal through natural methods such as afforestation, which absorb Carbon dioxide and decrease the residual emissions that are deemed unavoidable. The net-zero of this relates to the total emissions added and removed which end up cancelling each other out. Ensuring net-zero emissions is vital to the prevention of severe impacts on the environment such as water pollution which is a growing concern in the modern day due to the lack of environmental regulations subject to large corporations and developing countries. Through this sustainability can be achieved by resource efficiency via the promotion of recycling and waste reduction.

The UK has set out to achieve net-zero in the future through the use of making buildings more energy efficient. According to Great.gov.uk they state (Great.gov.uk, n.d) "The UK has set a target of net zero carbon emissions by 2050. Making buildings more energy efficient through modern construction methods and moving away from fossil-fuel boilers, are key to industry meeting this target." which supports the idea that the UK is attempting to battle carbon emissions in the future through using energy efficient buildings by implementing new architectural practices and trying to inspire innovation within the industrial architecture industry. The UK has also successfully identified the area they need to focus on to complete this target which is the movement away from fossil fuels. Moving away from fossil fuels is an essential part of the target's completion as fossil fuels have dangerous environmental impacts onto the environment which can cause the target of 2050 to be delayed and possibly never achieved if a reliable alternative to fossil fuels cannot be created or used realistically within the remaining 25 years until the deadline. According to (Bertrand, 2021) "Fossil fuels produce hazardous air pollutants, including sulfur dioxide and other particulate matter which are harmful to the

environment and human health" This is a highly valid comment as it has been scientifically backed and also comes from a reliable source being the Environmental and Energy Study Institute. We can begin to understand that fossil fuels have a large impact on our environment and show the possible long-term consequences of fossil fuel exposure and usage such as respiratory problems which will cause life expectancies to decrease on average and cause emergency services and NHS workers to be put under more stress to support these future issues. This long term effect of fossil fuels is an essential motivation for the UK government to achieve their goals by setting deadlines and stepping stones to achieve an overall goal of reducing carbon emissions and achieving net-zero.

The UK government, in the Climate Change Act 2008, declared they must show progress for this goal ([UK government, 2019](#)) "The government must report on progress towards meeting carbon budgets and the 2050 target at least every five years, making adjustments to policies as necessary." This suggests that the UK government is dedicated to showing their progress towards their commitment to net-zero emissions and are willing to change their policies in order to support this. However, from this declaration coming from the UK government itself it may cause some issues such as being a biased statement and unrealistic as they may not follow through with these updates or may not be truthful making this information less valid from its original state yet still fairly valid because it still from an official government source.

However the UK government has made several policies and acts as an attempt to control the damage to the environment and to support sustainability. The Environmental Act of 2021 is a perfect example of the UK's attempts to combat environmental damage. As defined by ClientEarth, ([ClientEarth, n.d](#)) "The Environment Act allows the UK to enshrine better environmental protection into law. It provides the Government with powers to set new binding targets, including for air quality, water, biodiversity, and waste reduction." As stated it provides the government with the ability to set new targets for different aspects of the environment from this the overall quality of sustainability will increase and also support the reduction of waste in industrial architecture as less unnecessary materials will be used which may be non-recyclable. This means that there are more opportunities in industrial architecture as the initial cost of buildings will be lowered due to waste reduction being implemented as a policy, this may attract architects to sustainable practices rather than non-sustainable methods.

What is the current state of sustainability in industrial architecture inside the UK?

After analysing past policies, defining the term 'sustainability' and discussing its important place in the world of industrial architecture, discussing the current state of sustainability inside of industrial architecture is essential to developing the possible solutions to the crisis of low sustainability within the UK.

Currently one of the main sources of carbon emissions, which affects overall sustainability, is buildings. According to a reliable architecture firm known as Firethorn Trust they state that ([Firethorn Trust, 2020](#)) "Buildings account for around 40% of global carbon emissions." This statistic shows how much of an impact architecture has on the environment on a global scale let alone just in the UK and accounts for all forms of architecture which shows the vast impact architects have on the carbon emissions of the UK which will directly affect the net-zero emissions goal for 2050 set by the UK government. This information from Firethorn Trust is essential to provide a spark towards a newer generation of architects to provide them the insight of how the state of sustainability currently is. Through this statistic we can begin to target certain areas of improvement to support us in the present day by analysing different aspects of the being and how they interact with the environment around us. Now that we understand the overall impact non-sustainable architecture has on the environment, globally, we can now look into more depth on the influence poor sustainability has on the UK individually. The UK has their own independent advisor on approaching climate change, known as the Climate Change Committee (CCE). Their role is to advise the UK government on how to tackle these issues revolving around climate change and report to parliament any progress, such as the goal of net-zero emissions for 2050. They also adapt for future issues relating to climate change and greenhouse gasses. The CCE also has information relating to the current greenhouse gas (GHG) emissions in the UK which allow the government to be able to determine whether they are on or off course of their targets set. The CCE has stated that ([Climate Change Committee, 2020](#)) "emissions from buildings were 87 Mt CO₂e in 2019, accounting for 17% of UK GHG emissions." From this

statistic we can see that 17% of all GHG emissions come from buildings which generate a total of 87 Mega tonnes of CO₂ equivalent meaning that it may not be CO₂ but it has been converted into the same amount of CO₂ it would have generated so we can see more clearly the impacts of GHG. Now understanding the amount of GHG that is produced through buildings, innovations must be produced in order to reduce these percentages. However, the information from this may be less accurate as this statistic was in 2019 being several years ago other factors may have changed these statistics such as new practices and adapting to these issues through the CCE. This links back to sustainability currently as the present can still be classed as 2019 because of the short-term effects that these statistics cause. Although it seems like several years ago the effects from this 17% is still present now and relevant to comprehending the current situation involving sustainable architecture.

Inside the UK it is clear that buildings are one the largest contributors to the carbon emissions through the statistics given by several architecture firms based all over the UK. from a reliable source ([Shubbar, Nasr, Fala, Al-Khafaji, 2021](#)) They state that "In the UK, the energy consumption of buildings is considerable, especially non-domestic buildings that account for nearly 35% of the greenhouse emissions" The non-domestic buildings in this statement refers to buildings that do not home a family for example, industrial buildings. This directly relates to industrial architecture as currently, 35% of greenhouse emissions are caused by these types of buildings meaning that there are plenty of improvements available in this industry. From understanding this statistic a better foundation is created to understanding how the UK can move away from old methods and practices to a more modern and sustainable way of thinking to provide the protection of species and future generations.

Sustainable Materials

Sustainable materials in the industrial sector of architecture are crucial to improve sustainability. Identifying potential solutions to this growing problem can promote new ideas . As a trusted architecture firm based in London stated. ([Mmoser, 2024](#)) "The only path to a carbon-neutral industry is shifting from a wasteful mindset of replacing old with new. Giving new life to fully depreciated carbon assets in developed countries and creating a framework to enable developing countries to build responsibly for the future" what Mmoser is trying to achieve from this statement is that providing support to smaller and developing countries can help them become more sustainably responsible when their industry grows and will not be impacted by old architectural practices with non-sustainable mindsets. Replacing the old with the new is important with industrial architecture as a way to advance to a new way of thinking and drift away from poorly sustainable structures such as our current industrial sector. Another architecture firm based in Hardwick Grange, Woolston is already thinking on how these advances could be made. ([Firethorn Trust, 2020](#)) "Using high-quality materials and insulation during construction is fundamental to achieving low energy consumption" these higher quality materials will prove to be crucial in the movement towards sustainable industrial architecture as it provides less energy being used per building and therefore improving the overall sustainability of industrial architecture. This can also be helped with improved insulation which will provide the building with less heat loss and less energy being used on appliances to keep the building warm. However using these higher quality materials does come at the drawback of having a higher initial cost of the building which may push industry owners away because of the higher cost upfront. Although this may be the case the overall cost of operations in the future will decrease as less energy gets consumed by the building per year meaning that the operational costs are lower than an average industrial building.

These sources are reliable in their validity however may be open to a potential bias due to both of these architecture firms, Mmoser and Firethorn Trust, being based inside of the UK meaning they have less of an outside perspective on the UK's sustainability concerns and may be unaware of how the UK's sustainability in industrial architecture differs from other countries. Further analysis into other academic studies may provide other viewpoints and increase the validity of these statements.

Currently, within industrial architecture brick is commonly used. Although this material may appear to be commonly used everywhere it does come with drawbacks and affects the sustainability of buildings. In this section of the chapter a deeper explanation of this material will be investigated. The average lifetime of this material is 150 years which shows why it is so commonly used because of its long lifespan. Due to this it is an extremely durable material being reliable to withstand extreme weather and keeping a building upright for a long time. Due to its resistance it can survive many

natural occurrences providing its use as a base material in constructing buildings, However because of this high durability, at the end of the bricks life cycle it ends up being dumped in landfills which pollute the soil around it causing damage to habitats and looking unpleasant to those that may pass by. Not only does this strongly harm the environment after its life but it also causes visual pollution. Especially in the future when these landfills build up and communities may have to live next to them. Brick also produces greenhouse gases due to the high energy usage when it is produced as it is produced with the support of high temperatures in factories.

In addition to the common uses of brick there is another material commonly used within industrial architecture. This material is metal, more specifically, aluminium. This metal contains a lifespan of roughly 75 years which is a very long lifespan compared to most materials. This lifespan proves aluminium's high durability. Properties of this metal also include being a good conductor of heat and electricity meaning it can easily transfer heat and electricity to certain places and is also a cheap product. A major environmental benefit of this specific metal is that on average 95% of it can be recovered through recycling meaning it is a very sustainable material because of its recyclability. However, a product of aluminium refining is a substance called 'red mud' which is highly toxic and dangerous which can cause damage to people and the environment if not disposed of carefully.

Another material usually used in industrial architecture is concrete. This material is more commonly used in everyday life than just industrial architecture however it still remains a large material used. Its lifespan ranges depending on how it is looked after but on average it can be anywhere from 50 to 100 years. Because concrete has extremely high durability it can withstand many types of deterioration during its lifetime but from this it can cause soil erosion and cause damage to the fertile layer of the earth which damages environmental growth and possible wildlife habitats. Also, a noticeable problem with concrete is how poorly energy efficient it is to use. Because of its transport it causes high amounts of energy consumption to move this material to different locations making it a poor material to keep using. The main issue with concrete is its damage to the ground and also its low energy efficiency when transporting. The energy efficiency could be increased by inventing new ways of transporting concrete from different locations such as using higher capacity vehicles to reduce carbon emissions per journey. However this will come at the cost of increasing the individual carbon footprint of each vehicle as it will require more energy to transport a higher volume of concrete in one journey. According to the science museum which is a valid source of information stated that ([The Science Museum, 2021](#)) "Cement production alone generates around 2.5 billion tonnes of carbon dioxide per year" which shows how much the production of cement generates. Cement is a material used to create concrete which causes high amounts of CO₂ into our atmosphere and damages many parts of the environment.

The next material that is commonly used is glass. Glass is used in buildings to provide a visual element to the occupants by allowing them to see outside while also protecting them from the weather such as sun, wind and rain. There are three main types of glass, single pane, double pane and triple pane. Single pane glass can last around 20 years which is a very small lifetime compared to other materials that have been evaluated so far. Double and triple pane glass have a larger lifespan due to this type of glass having a higher durability and can last longer being up to 40 years of life which is double that of single pane glass. This makes double and triple pane glass a lot more useful to implement into a building as it provides a more durable and stable building along with its longer lifetime. Glass is a recyclable material which makes it a fairly sustainable material compared to the other three, meaning it can be reused in different situations. To reduce energy consumption higher pane glass is a more viable option as it is a good insulator for heat so it can trap heat as heat cannot go through the glass. The production of glass is a sustainable method compared to other processes because glass consists of sand and lime stone which are natural materials and can be found in the world meaning little energy is used to collect these materials. However, the production of glass also causes CO₂ emissions ([University park, 2023](#)) "Worldwide, glass manufacturing produces at least 86 million tons of carbon dioxide every year" These emissions, although very high, are a significant decrease compared to materials such as concrete. This statistic is from a university in Penn State which is based in America which gives an outside perspective on CO₂ emissions but may be less reliable as it is only a university and may be not experts in this field.

To conclude this section, the impact of extraction materials is vital in showing the environmental impacts. Mentioned by ([Lombera, Roj, 2010](#)) "the construction industry sector consumes up to 60% of all raw materials extracted from the Earth. " This shows the importance of finding better alternatives to

these finite resources that may eventually be used up in an attempt to reduce the extraction of raw materials which takes up a lot of energy and reduces the finite materials of Earth.

Conclusion

To conclude, improving sustainability in industrial architecture is more than just focusing on the building but also its overall impact on society and the environment. From this essay a new understanding of architectural practices can be invented through adaptation.

Improving sustainability can be achieved through changing materials to more sustainable alternatives in an attempt of having less of an impact on the environment and reducing the energy used to process and transport materials so that more can be produced while using the same amount of energy which will improve efficiency. Upgrading glass to double glazed can support architects in providing a more sustainable building as double glazed windows trap more heat and reduce the heat loss of a building which will increase the sustainability.

Spreading awareness in communities of the importance of sustainability may also help improve industrial architecture's sustainability. Promotion of sustainability inside of communities will raise the quality of life and therefore show the benefits of sustainable architecture. This new-found awareness will provide not only a more sustainable industrial architecture but a cleaner society as it will make individuals more aware of their actions meaning GHG will reduce even further from this and go down generations as they will be brought up in a clean society. Spreading awareness could also include informing society on updates within sustainability and the current state of it. Doing so will supply society with the knowledge to do their part by changing their ideas on sustainability.

New policies would benefit sustainability as it will give architecture firms new goals to reach and enforce the overall concept of sustainability. Enforcing these policies by including repercussions for those who do not meet the requirements will also yield higher results of a reduction in GHG and poor sustainable buildings. These repercussions are important as it was suggested that the UK wants to move away from fossil fuels yet (Firethorn Trust, 2020) states "Buildings account for around 40% of global carbon emissions." meaning that a sustainable method has still not been achieved to support the UK's goals.

Ultimately, sustainable industrial architecture is not only achievable but necessary. Through the use of education, new political policies and innovations in practices a better environment can be created supporting not only society but the planet as a whole.

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